The Symcoder Encoder: Methodology and Correctness

A pedagogical account of the permutation-invariant embedding algorithm

Contents

1 Overview

The encoder takes as input a *physics event*: a finite collection of particle vectors grouped by particle type, together with a set of scalar operations. It produces a fixed-length real vector that is invariant under any permutation of particles of the same type, and that contains enough information to separate generic events.

The construction proceeds in two phases.

1. **Phase 1 — single-orbit encoding.** For each distinct *FlavouredOperator* (an operation with a fixed recipe for drawing argument labels from the particle groups), evaluate the operation on every valid label combination and sort the results. This yields a permutation-invariant sequence of real numbers describing the marginal distribution of each scalar feature.
2. **Phase 2 — pair-association encoding.** For each pair of FlavouredOperators, evaluate the complex number $z_k = \mathrm{eval}(u_k) + i\,\mathrm{eval}(v_k)$ for every pair (u_k, v_k) of atoms that share a specific pattern of label overlap (*one association*). Embed the resulting multiset into permutation-invariant polynomial coefficients. Several optimisations reduce storage without losing information.

The two phases together form a complete, permutation-invariant encoding. Phase 1 captures individual feature statistics; Phase 2 captures pairwise correlations between features.

2 Mathematical Setup

2.1 Events, Groups, and Labels

Definition 2.1 (Vector group)**.** A *vector group* $\mathcal{V} = (\mathcal{L}, d)$ consists of a finite label set $\mathcal{L} = \{\ell_1, \ldots, \ell_n\}$ and a dimension $d \in \mathbb{N}$. The *size* of the group is $|\mathcal{L}| = n$.

Definition 2.2 (Context)**.** A *context* $\mathcal{C} = (\mathcal{V}_1, \ldots, \mathcal{V}_m)$ is an ordered tuple of vector groups with disjoint label sets. Let $n_i = |\mathcal{V}_i|$ be the size of group i.

Definition 2.3 (Event)**.** An *event* E is an assignment $\ell \mapsto \mathbf{v}_\ell \in \mathbb{R}^d$ of a d-dimensional vector to every label ℓ across all groups.

The symmetry group acting on events is the product of symmetric groups

$$G = \mathfrak{S}_{n_1} \times \cdots \times \mathfrak{S}_{n_m},$$

where each $\mathfrak{S}_{n_i}$ permutes the labels of group i. Two events are *physically equivalent* if one is obtained from the other by a G-action, i.e. by relabelling particles within each type. The encoder must produce the same output for physically equivalent events.

2.2 Operations

Definition 2.4 (Operation)**.** An *operation* op is characterised by:

- a *rank* $r \in \mathbb{N}$ (number of vector arguments),
- a *parity* $p \in \{+1, -1\}$,
- an *argument symmetry* $\sigma \in \{\textsc{Symmetric}, \textsc{Antisymmetric}\}$, and
- an *evaluation function* $op_{fn} : (\mathbb{R}^d)^r \to \mathbb{R}$.

For Symmetric operations, op_{fn} is invariant under all permutations of its r arguments. For Antisymmetric operations, op_{fn} changes sign under any transposition of two arguments.

Example 2.5. The dot product $dot(\mathbf{u}, \mathbf{v}) = \mathbf{u} \cdot \mathbf{v}$ has rank 2, parity $+1$, and is SYMMETRIC.

The triple product $\varepsilon_3(\mathbf{u}, \mathbf{v}, \mathbf{w}) = \mathbf{u} \cdot (\mathbf{v} \times \mathbf{w})$ has rank 3, parity -1, and is ANTISYMMETRIC.

2.3 Atoms

An *atom* is an operation applied to a specific ordered tuple of labels. Because the argument symmetry of the operation constrains how permutations of the labels affect the result, each unordered combination of labels gives rise to either one or two atoms.

Definition 2.6 (Atom)**.** Let *op* be an operation of rank r and let $(\ell_1, \ldots, \ell_r)$ be a tuple of distinct labels in canonical (lexicographic) order within each group. An *atom* is a triple

$$u = (op,\ (\ell_1, \ldots, \ell_r),\ \varepsilon)$$

where $\varepsilon \in \{+1, -1\}$ is the *sign*.

- For SYMMETRIC *op*: only $\varepsilon = +1$ arises.
- For ANTISYMMETRIC *op*: both $\varepsilon = +1$ and $\varepsilon = -1$ arise, one for each canonical label tuple. The sign ε records the parity of the permutation that maps the canonical ordering to a particular chosen reference ordering.

Definition 2.7 (Evaluation)**.** The evaluation of atom $u = (op, (\ell_1, \ldots, \ell_r), \varepsilon)$ on event E is

$$\mathrm{eval}(u, E) = \varepsilon \cdot op_{fn}(\mathbf{v}_{\ell_1}, \ldots, \mathbf{v}_{\ell_r}).$$

The sign ε means that for an ANTISYMMETRIC operation the two atoms sharing the same label tuple satisfy $\mathrm{eval}(u^+, E) = -\mathrm{eval}(u^-, E)$ for every event E, by antisymmetry of op_{fn}.

2.4 Flavours and FlavouredOperators

Atoms are organised by how many labels each argument draws from each group.

Definition 2.8 (Flavour)**.** A *flavour* for a context with m groups is a tuple $F = (k_1, \ldots, k_m)$ of non-negative integers with $\sum_i k_i = r$ (the rank of the associated operation). We require $k_i \le n_i$ for all i.

Definition 2.9 (FlavouredOperator)**.** A *FlavouredOperator* $\mathrm{FO}(op, F)$ is the set of all atoms whose operation is *op* and whose label tuple draws exactly k_i labels from group i, for each i.

The number of atoms in a FlavouredOperator is

$$|\mathrm{FO}(op, F)| = \begin{cases} \displaystyle\prod_{i=1}^{m} \binom{n_i}{k_i} & \text{if } op \text{ is SYMMETRIC,} \\ \displaystyle 2\prod_{i=1}^{m} \binom{n_i}{k_i} & \text{if } op \text{ is ANTISYMMETRIC.} \end{cases} \tag{1}$$

The factor of 2 for ANTISYMMETRIC operations reflects the two sign choices per label combination.

Definition 2.10 (repL)**.** For a context $\mathcal{C}$ and a set of operations $\mathcal{K}$, the *repL* is the list of all valid FlavouredOperators:

$$\mathrm{repL}(\mathcal{C}, \mathcal{K}) = \big\{\mathrm{FO}(op, F) : op \in \mathcal{K},\ F \text{ is a valid flavour for } op \text{ and } \mathcal{C}\big\}.$$

Every atom in the encoding vocabulary appears in exactly one FlavouredOperator.

3 Phase 1: Single-Orbit Encoding

The first phase encodes each FlavouredOperator's evaluation values as a permutation-invariant sequence of reals.

3.1 The Symmetric Case

For a Symmetric FlavouredOperator $fo = \mathrm{FO}(op, F)$, let $n = \prod_i \binom{n_i}{k_i}$ be the number of atoms (all with sign $+1$). Evaluating on event E gives n real numbers

$$a_1 = \mathrm{eval}(u_1, E), \quad a_2 = \mathrm{eval}(u_2, E), \quad \ldots, \quad a_n = \mathrm{eval}(u_n, E).$$

The **Phase 1 encoding** of fo is the sorted sequence

$$\big(a_{(1)}, a_{(2)}, \ldots, a_{(n)}\big), \qquad a_{(1)} \le a_{(2)} \le \cdots \le a_{(n)}.$$

Sorting is the canonical permutation-invariant operation. The sorted sequence is a complete encoding of the multiset $\{a_1, \ldots, a_n\}$, and hence of all symmetric functions of that multiset.

3.2 The Antisymmetric Case: Sign Compression

For an Antisymmetric fo, the atom set contains pairs (u^+, u^-) for each label combination, with $\mathrm{eval}(u^+, E) = -\mathrm{eval}(u^-, E)$. The full set of $2n$ evaluation values is therefore

$$\{a_1, -a_1, a_2, -a_2, \ldots, a_n, -a_n\}, \qquad \text{where } a_j = \mathrm{eval}(u_j^+, E).$$

This multiset is antisymmetric about zero: knowing the positive half $\{a_1, \ldots, a_n\}$ (or equivalently the absolute values $\{|a_j|\}$) determines the full multiset.

Proposition 3.1 (Sign compression)**.** *The multiset $\{|a_1|, \ldots, |a_n|\}$ is invariant under the action of G.*

Proof. A group element $g \in G$ permutes atoms and may change the canonical label ordering, thereby flipping the sign of some atoms (i.e. swapping $u_j^+ \leftrightarrow u_j^-$ for certain j). However, $|a_j| = |\mathrm{eval}(u_j^+, E)| = |\mathrm{eval}(u_j^-, E)|$ is invariant under sign flip, so the multiset $\{|a_j|\}$ is unchanged. □

The decision to apply sign compression is made from the operation's argument symmetry alone — never from the numerical values of a_j.

Phase 1 encoding (antisymmetric case): store the sorted sequence

$$\big(|a|_{(1)}, |a|_{(2)}, \ldots, |a|_{(n)}\big), \qquad |a|_{(1)} \le \cdots \le |a|_{(n)},$$

contributing $n = \frac{1}{2}|fo|$ reals instead of $2n$.

3.3 Summary

Each distinct FlavouredOperator fo contributes

$$\ell_1(fo) = \begin{cases} |fo| & \text{if Symmetric}, \\ \frac{1}{2}|fo| & \text{if Antisymmetric}, \end{cases} \tag{2}$$

reals to Phase 1. In both cases $\ell_1(fo) = \prod_i \binom{n_i}{k_i}$, the number of distinct label combinations.

4 Phase 2: Pair-Association Encoding

Phase 2 encodes pairwise correlations between atoms from (possibly different) FlavouredOperators. The central objects are *PairFlavours*, which classify atom-pairs by orbit type under G.

4.1 PairFlavours and Overlap Blocks

Definition 4.1 (PairFlavour)**.** A *PairFlavour* is a tuple

$$\mathrm{pf} = (op_u, F_u, op_v, F_v, s),$$

where (op_u, F_u) and (op_v, F_v) are the operation-flavour pairs of atoms u and v, and $s = (s_1, \ldots, s_m)$ is the *overlap* vector: s_i is the number of labels shared between u and v from group i. We require $0 \leq s_i \leq \min(k_{u,i}, k_{v,i})$ for all i.

By convention the two sides are canonically ordered so that $(op_u, F_u) \leq (op_v, F_v)$ under a fixed total order; this ensures that $\mathrm{pf}(u, v) = \mathrm{pf}(v, u)$.

Proposition 4.2. *Two atom-pairs (u, v) and (u', v') lie in the same G-orbit if and only if they have the same PairFlavour.*

Definition 4.3 (Overlap block)**.** An *overlap block* is the set of all PairFlavours sharing the same (op_u, F_u, op_v, F_v), differing only in the overlap s. Each member of an overlap block is called an *association.*

The valid overlap range for group i is

$$s_i \in \big[\max(0,\ k_{u,i} + k_{v,i} - n_i),\ \min(k_{u,i}, k_{v,i})\big].$$

4.2 The Association Table

Fix an overlap block $B = \{(op_u, F_u, op_v, F_v, s) : s \text{ valid}\}$. Imagine a matrix whose columns are labelled by the atoms of $\mathrm{FO}(op_u, F_u)$ and whose rows are labelled by the atoms of $\mathrm{FO}(op_v, F_v)$. Each cell (u_j, v_k) belongs to exactly one association: the one whose overlap s_i equals the number of labels shared by u_j and v_k from group i.

Example 4.4. Let $n = 3$ (labels a, b, c), one group, and $op = dot$ with rank 2 and flavour $F = (2)$. The atoms are $dot(a, b)$, $dot(a, c)$, $dot(b, c)$. The self-pairing overlap block $(dot, (2), dot, (2))$ has associations $s \in \{1, 2\}$ (overlap $s = 0$ is impossible here since $k_u + k_v - n = 2 + 2 - 3 = 1$).

	$dot(a,b)$	$dot(a,c)$	$dot(b,c)$
$dot(a,b)$	$s=2$	$s=1$	$s=1$
$dot(a,c)$	$s=1$	$s=2$	$s=1$
$dot(b,c)$	$s=1$	$s=1$	$s=2$

Association $s = 2$ (diagonal): 3 cells; association $s = 1$ (off-diagonal): 6 cells.

The number of atom-pairs in association pf is

$$\mathrm{count}(\mathrm{pf}) = \prod_{i=1}^{m} \binom{n_i}{s_i} \binom{n_i - s_i}{k_{u,i} - s_i} \binom{n_i - k_{u,i}}{k_{v,i} - s_i}. \tag{3}$$

The three binomial factors count: (i) the shared labels, (ii) the extra u-labels from the remaining pool, (iii) the extra v-labels from what is left.

Remark 4.5. The cells of the association table partition exactly into the associations of the block:

$$|\mathrm{FO}(op_u, F_u)| \cdot |\mathrm{FO}(op_v, F_v)| = \sum_{\mathrm{pf} \in B} \mathrm{count}(\mathrm{pf})$$

(counting sign variants on both sides for antisymmetric operations).

4.3 Dropping Self-Pairing Associations (`NULL_SELF`)

A *self-pairing association* is one in which $op_u = op_v$, $F_u = F_v$, and the overlap equals the maximum: $s_i = k_{u,i}$ for all i. Under this condition every atom-pair (u_j, v_k) satisfies $u_j = v_k$, so

$$z_j = \mathrm{eval}(u_j, E) + i\,\mathrm{eval}(v_j, E) = a_j + i\,a_j = (1+i)\,a_j,$$

where $a_j = \mathrm{eval}(u_j, E)$ is already stored in the Phase 1 orbit for $fo(op_u, F_u)$. The entire z-multiset of the association is therefore $\{(1+i)\,a_j\}$, which is determined by Phase 1 alone.

All self-pairing associations are dropped from the encoding regardless of their count count(pf). They are labelled `NULL_SELF` in the output metadata and contribute zero reals.

Example 4.6. In Example 4.4, the association $s = 2$ is the self-pairing one (diagonal cells of the association table). Its three z-values equal $(1+i)\,dot(a, b, E)$, $(1+i)\,dot(a, c, E)$, $(1+i)\,dot(b, c, E)$ — identical to $(1 + i)$ times the Phase 1 orbit for $fo(dot, (2))$. It is therefore always dropped as `NULL_SELF`, leaving only the $s = 1$ association in the block.

4.4 Dropping the Largest Remaining Association (`NULL_COMP`)

After removing all self-pairing associations, the association with the largest count(pf) among those *remaining* in each overlap block is not encoded explicitly. Instead, it is recovered by the decoder via a multiset-complementation argument.

Self-pairing associations are excluded before this step so that the complementation drop removes the largest thing that cannot already be recovered from Phase 1. (If the self-pairing association happened to be the largest, a naïve largest-first policy would waste the complementation drop on it, leaving a genuine non-trivial association encoded redundantly.)

Let B' denote the non-self-pairing associations of block B, and let $\mathrm{pf}^* \in B'$ be the one with the largest count. Write

$$U = \{\mathrm{eval}(u_j, E) : u_j \in \mathrm{FO}(op_u, F_u)\} \quad \text{and} \quad V = \{\mathrm{eval}(v_k, E) : v_k \in \mathrm{FO}(op_v, F_v)\}$$

for the multisets of evaluation values (known from Phase 1 encodings of $\mathrm{FO}(op_u, F_u)$ and $\mathrm{FO}(op_v, F_v)$).

The multiset of complex numbers

$$Z = \{z_{jk} = \mathrm{eval}(u_j, E) + i\,\mathrm{eval}(v_k, E) : (u_j, v_k) \text{ any pair with } u_j \in \mathrm{FO}_u,\ v_k \in \mathrm{FO}_v\}$$

is the multiset product $U \oplus_{\mathbb{C}} V = \{a + ib : a \in U, b \in V\}$, which is determined entirely by U and V. The associations of block B partition Z into disjoint submultisets Z_{pf}. The self-pairing associations $Z_{\mathrm{pf}_{\mathrm{self}}}$ are already known from Phase 1 (see section 4.3), so knowing all non-self-pairing associations except pf^* determines Z_{pf^*} by complement:

$$Z_{\mathrm{pf}^*} = Z \setminus \Big(\bigsqcup_{\mathrm{pf} \in B_{\mathrm{self}}} Z_{\mathrm{pf}} \sqcup \bigsqcup_{\mathrm{pf} \in B',\, \mathrm{pf} \neq \mathrm{pf}^*} Z_{\mathrm{pf}} \Big).$$

Proposition 4.7 (Complementation is exact). *The above complementation works for* all *events E, including those where some evaluation values coincide. Since Z is a multiset, repeated values appear with the correct multiplicity, and the complement operation on multisets accounts for this correctly.*

Remark 4.8 (Choosing which association to drop). The complementation argument is valid for any choice of $\mathrm{pf}^* \in B'$. Choosing the largest among B' minimises storage.

Remark 4.9 (Single-association blocks). If an overlap block has only one non-self-pairing association (which occurs when $\mathrm{FO}(op_u, F_u)$ and $\mathrm{FO}(op_v, F_v)$ draw labels from completely disjoint groups, so $s = (0, \ldots, 0)$ is the only non-self valid overlap), then that association is the unique pf^* and is dropped as `NULL_COMP`. No information is lost. If *all* associations in a block are self-pairing (possible only when FO_u and FO_v use every label in each group, so $k_i = n_i$), the entire block has no non-self associations and no `NULL_COMP` is emitted; the block contributes nothing to the output.

4.5 The Z-value Embedding

For each surviving association pf (i.e. each non-dropped member of each overlap block), the encoder computes a set of complex numbers and then produces a permutation-invariant polynomial encoding.

4.5.1 The positive-sign subset

For each atom-pair (u_j, v_k) with both $u_j.\varepsilon = +1$ and $v_k.\varepsilon = +1$, define

$$z_k = \mathrm{eval}(u_j, E) + i\,\mathrm{eval}(v_k, E).$$

This subset has exactly $\mathrm{count}(\mathrm{pf})$ elements regardless of which operations are antisymmetric, because for each base label combination there is exactly one (u^+, v^+) pair.

4.5.2 Permutation-invariant embedding

The multiset $\{z_k\}$ (of size $n = \mathrm{count}(\mathrm{pf})$) is embedded by the Echinocoder polynomial encoder: compute the monic polynomial whose roots are $\{-z_k\}$, and return all coefficients except the leading (monic) one, ordered by descending degree. This yields n complex numbers that are invariant under any permutation of the z_k values (i.e. under any relabelling of the pairs).

Each complex coefficient is unpacked into two reals $(\mathrm{Re}, \mathrm{Im})$, contributing $2n = 2\mathrm{count}(\mathrm{pf})$ reals in total.

4.6 Sign-Symmetry Compression of the Embedding

When one or both operations in an association are Antisymmetric, the full orbit of z-values (over all sign combinations of u and v) has a forced structure that can be exploited to halve (or quarter) the apparent cost of embedding the full orbit.

The key observation is that the full orbit's z-values come in symmetry-related groups:

- Sym $\times$ Antisym (SA): the full orbit is $\{z_k, \overline{z}_k\}$ — conjugate-closed.
- Antisym $\times$ Sym (AS): the full orbit is $\{z_k, -\overline{z}_k\}$ — anti-conjugate-closed.
- Antisym $\times$ Antisym (AA): the full orbit contains $\{z_k, \overline{z}_k, -z_k, -\overline{z}_k\}$, so the squares $\{z_k^2\}$ form a conjugate-closed set of size n.

In each non-trivial case, the characteristic polynomial of the full orbit has forced coefficient structure (real coefficients, or alternating real/imaginary, etc.) that allows $2n$ reals to encode the same information as the brute-force $4n$ or $8n$ reals. The details are as follows.

4.6.1 SS: Symmetric × Symmetric

Embed $\{z_k\}$ (n complex roots) directly. Output: n complex $= 2n$ reals. No forced structure.

4.6.2 SA: Symmetric × Antisymmetric

The full orbit is $\{z_k, \overline{z}_k\}$ ($2n$ roots). The corresponding polynomial has *real* coefficients (a conjugate-closed root set implies real coefficients). Output: $2n$ real coefficients, packed as n complex numbers ($2n$ reals total).

4.6.3 AS: Antisymmetric × Symmetric

The full orbit is $\{z_k, -\overline{z}_k\}$ ($2n$ roots). The polynomial has even-degree coefficients real and odd-degree coefficients purely imaginary. Output: n independent reals of each parity, again packed as n complex numbers ($2n$ reals).

4.6.4 AA: Antisymmetric × Antisymmetric

The full orbit is $\{z_k, \overline{z}_k, -z_k, -\overline{z}_k\}$ ($4n$ roots in n quadruples). Setting $w_k = z_k^2$ reduces to an SA-type structure in the variable w: the multiset $\{w_k, \overline{w}_k\}$ is conjugate-closed and its polynomial has real coefficients. Output: n complex $= 2n$ reals, after embedding $\{w_k^2\}$ as in SA.

Proposition 4.10 (Output size)**.** *Every association, regardless of symmetry class, contributes exactly* $2\mathrm{count}(\mathrm{pf})$ *reals to the Phase 2 encoding.*

The compression is in avoiding the naive approach of embedding the full orbit (which would require $2 \times \mathrm{orbit\_size}(\mathrm{pf})$ reals and grows by a factor of 2 or 4 for antisymmetric operations).

5 Correctness

5.1 Permutation Invariance

Theorem 5.1. *The complete encoding (Phase 1 + Phase 2) is invariant under the action of the symmetry group* $G = \mathfrak{S}_{n_1} \times \cdots \times \mathfrak{S}_{n_m}$.

Proof sketch. **Phase 1.** A group element $g \in G$ permutes atoms within each FlavouredOperator, and for ANTISYMMETRIC operations may flip the sign of some atoms. The SYMMETRIC Phase 1 encoding sorts evaluation values, and sorting commutes with any permutation of the values. The ANTISYMMETRIC encoding sorts absolute values, which are additionally invariant under sign flips (section 3.2).

Phase 2. The complex numbers $z_k = \mathrm{eval}(u_j, E) + i\,\mathrm{eval}(v_k, E)$ are defined for pairs (u_j, v_k) in the positive-sign subset of association pf. Under g, the set of such pairs is permuted (with possible sign changes, handled by the 5a symmetry classes in section 4.6). The polynomial embedding of a multiset is invariant under any permutation of its elements, so the embedding is G-invariant. The complementation step (**??**) produces the same dropped-association values because it operates on the multiset Z_{pf^*}, which is the set-theoretic complement of the stored associations within $Z = U \oplus_{\mathbb{C}} V$, and Z, U, V are all G-invariant. □

5.2 Information Completeness

The encoding is *complete* if the full set of evaluation values for all atoms and atom-pairs can be reconstructed from the encoded vector. Completeness is required for the encoder to separate generic events.

Theorem 5.2 (Phase 1 completeness)**.** *The Phase 1 encoding of* $\mathrm{FO}(op, F)$ *is a complete encoding of the multiset* $\{\mathrm{eval}(u, E) : u \in \mathrm{FO}\}$*.*

Proof. The sorted sequence is the canonical representative of the multiset; it uniquely determines the multiset. For the ANTISYMMETRIC case, the sorted absolute values $\{|a_j|\}$ determine the multiset $\{a_j, -a_j\}$ because each absolute value contributes exactly one positive/negative pair. □

Theorem 5.3 (Phase 2 completeness)**.** *Given the Phase 1 encodings of all FlavouredOperators, the Phase 2 encoding (after complementation) is complete.*

Proof. For each overlap block B:

1. The stored associations' z-multisets $\{Z_{\mathrm{pf}}\}_{\mathrm{pf}\in\mathrm{ASSOC}}$ are recovered from their polynomial embeddings (which are injective on multisets by the Echinocoder construction).
2. The Phase 1 encodings supply the multisets U and V, hence $Z = U \oplus_{\mathbb{C}} V$.
3. Self-pairing associations (`NULL_SELF`) satisfy $Z_{\mathrm{pf}_{\mathrm{self}}} = \{(1+i)\, a_j\}$, reconstructible from Phase 1.
4. The complementation-dropped association (`NULL_COMP`) satisfies $Z_{\mathrm{pf}^*} = Z \setminus \big(\bigsqcup_{\mathrm{pf}\in B_{\mathrm{self}}} Z_{\mathrm{pf}} \sqcup \bigsqcup_{\mathrm{pf}\in\mathrm{ASSOC}} Z_{\mathrm{pf}}\big)$, uniquely determined.
5. The sign-symmetry compression is reversible: the full orbit z-multiset is reconstructed from the positive-sign subset and the symmetry class of the association.

□

Remark 5.4 (Degenerate events)*.* Completeness holds for *all* events, including degenerate ones where some particles have identical vectors. Identical particles produce repeated evaluation values; these are handled correctly as multiset multiplicities throughout (in Phase 1 sorted sequences, in $Z = U \oplus_{\mathbb{C}} V$, and in the multiset complement). Two numerically identical particles are indistinguishable, which is the physically correct behaviour for a permutation-invariant encoder.

6 Output Structure

The output of the encoder for a given plan $(\mathcal{C}, \mathcal{K})$ and event E is a concatenation of real-valued segments. The structure depends only on the plan; it is the same for every event.

6.1 Segment types

`ORBIT` One per distinct FlavouredOperator *fo* in $\mathrm{repL}(\mathcal{C}, \mathcal{K})$. Length: $\prod_i \binom{n_i}{k_i}$ reals (*sign-compressed* for ANTISYMMETRIC operations; label `sign_compressed=True` in the metadata).

`ASSOC` One per stored association (every non-dropped member of each overlap block). Length: $2\mathrm{count}(\mathrm{pf})$ reals.

`NULL_SELF` One per self-pairing association (dropped because its z-values equal $(1+i)\cdot a_j$ and are deducible from Phase 1; see section 4.3). Length: 0 reals.

`NULL_COMP` One per complementation-dropped association (largest non-self-pair per block; see section 4.4). Length: 0 reals.

Both `NULL_SELF` and `NULL_COMP` entries appear at their logical position in the overlap block sequence so a decoder can reconstruct which association was omitted and by which argument.

All `ORBIT` segments precede all `ASSOC`/`NULL_SELF`/`NULL_COMP` segments. Within Phase 2, segments are grouped by overlap block, ordered by a canonical sort key $(op_u\text{-key}, op_v\text{-key}, s)$ (where op-key $= (\text{name}, r, p, \sigma, F)$), with overlap s ascending lexicographically within each block.

6.2 Total output length

$$L = \underbrace{\sum_{fo \in \text{repL}} \prod_i \binom{n_i}{k_i}}_{\text{Phase 1}} + \underbrace{\sum_{\substack{\text{pf} \in \text{all blocks} \\ \text{pf} \notin B_{\text{self}} \\ \text{pf} \neq \text{pf}^*_B}} 2\,\text{count}(\text{pf})}_{\text{Phase 2}}, \tag{4}$$

where B_{self} denotes the set of self-pairing associations in block B and pf^*_B denotes the largest non-self-pairing association (the `NULL_COMP`) in block B.

6.3 Human-readable metadata

The function `describe_encoding(plan)` returns a list of `SegmentInfo` records, one per segment, containing: segment kind (`ORBIT`/`ASSOC`/`NULL_SELF`/`NULL_COMP`), start index, length, operation names, flavours, overlap, symmetry class (`SS`/`SA`/`AS`/`AA`), and the `sign_compressed` flag for `ORBIT` segments. This list is a complete description of the encoder's output structure given only the plan — no event data is needed.

7 Algorithm Summary

Step	Input	Output
Construct `repL`	plan $(\mathcal{C}, \mathcal{K})$	list of FlavouredOperators
Compute PairFlavours	`repL`	all valid pf, grouped into overlap blocks
Phase 1 per fo	`fo.atoms()`, event E	$\ell_1(fo)$ sorted reals
Phase 2: drop self-pairs (`NULL_SELF`)	overlap blocks	self-pair map
Phase 2: choose pf* per block (`NULL_COMP`)	remaining assocs	dropped-association map
Phase 2: z-values per pf $\neq$ pf*	atom pairs, event E	count(pf) complex numbers
Phase 2: embed z-values (5a)	$\{z_k\}$, symmetry class	count(pf) complex coefficients
Phase 2: unpack to reals	complex coefficients	2count(pf) reals
Concatenate	all segments	output vector of length L

A Notation Reference

Symbol	Meaning
m	number of particle groups in the context
n_i	size of group i (number of labels)
$G = \mathfrak{S}_{n_1} \times \cdots \times \mathfrak{S}_{n_m}$	symmetry group
r	rank of an operation (number of vector arguments)
$F = (k_1, \ldots, k_m)$	flavour: label counts per group, $\sum k_i = r$
$\mathrm{FO}(op, F)$	FlavouredOperator
$\varepsilon \in \{+1, -1\}$	sign of an atom
$\mathrm{eval}(u, E)$	scalar evaluation of atom u on event E
$\mathrm{count}(\mathrm{pf})$	number of atom-pairs in association pf
$s = (s_1, \ldots, s_m)$	overlap vector of a PairFlavour
Z_{pf}	multiset of z-values for association pf
$U \oplus_{\mathbb{C}} V$	complex Cartesian sum $\{a + ib : a \in U, b \in V\}$